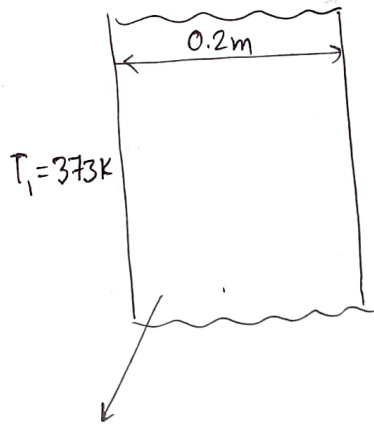


(A) Transient Heat Conduction through a plane wall :- using explicit method

(i) Numerical Solution :-



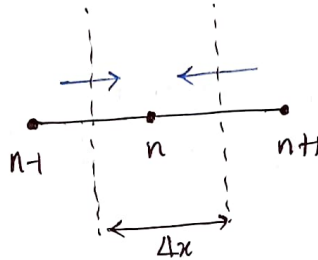
$$T_a = 303K$$

$$h = 45 \frac{W}{m^2K}$$

(a) for 1st node $n=1$:-

$$T_1^i = 373K \text{ (for all time } T_1 = 373K)$$

(b) for inner nodes $n=2, 3, 4, \dots, N-1$



$$\frac{KA(T_{n-1} - T_n)}{\Delta x} + \frac{KA(T_{n+1} - T_n)}{\Delta x} = \rho A \Delta x C_p \left(\frac{T_n^{i+1} - T_n^i}{\Delta t} \right)$$

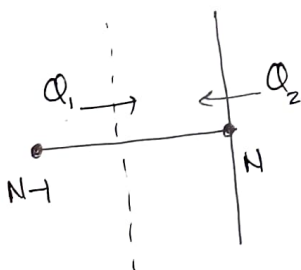
$$T_{n+1}^i + T_{n-1}^i - 2T_n^i = \frac{\Delta x^2}{\alpha \Delta t} (T_n^{i+1} - T_n^i)$$

taking $\frac{\Delta x^2}{\alpha \Delta t}$ as $\frac{1}{\tau}$ where τ = Fourier no.

$$T_{n+1}^i + T_{n-1}^i - 2T_n^i = \frac{T_n^{i+1} - T_n^i}{\tau}$$

$$T_n^{i+1} = \tau (T_{n+1}^i + T_{n-1}^i) + (1 - 2\tau) T_n^i \quad *$$

(c) for last node $n=N$.



$$KA \left(\frac{T_{N-1}^i - T_N^i}{\Delta x} \right) + hA(T_\infty - T_N^i) = \rho A \frac{\Delta x}{2} C_p \left(\frac{T_N^{i+1} - T_N^i}{\Delta t} \right)$$

$$T_{N-1}^i - T_N^i + \frac{h\Delta x}{K} (T_\infty - T_N^i) = \frac{\rho \Delta x^2 C_p}{2K} \left(\frac{T_N^{i+1} - T_N^i}{\Delta t} \right)$$

$$2\tau (T_{N-1}^i - T_N^i) + 2\tau \left(\frac{h\Delta x}{K} (T_\infty - T_N^i) \right) = T_N^{i+1} - T_N^i$$

$$T_N^{i+1} = T_N^i \left(1 - 2\tau - 2\tau \frac{h\Delta x}{K} \right) + 2\tau T_{N-1}^i + \frac{2\tau h\Delta x T_\infty}{K}$$

→ Stability criteria :-

$$1 - 2\tau - 2\tau \frac{h\Delta x}{K} \geq 0$$

$$\frac{h\Delta x}{K} \leq \frac{1 - 2\tau}{2\tau} = \frac{1}{2\tau} - 1$$

$$\frac{1}{2\tau} \geq \frac{h\Delta x + K}{K}$$

$$\tau \leq \frac{K}{2(h\Delta x + K)}$$

$$\Delta t \leq \frac{K}{2(h\Delta x + K)} \times \frac{\Delta x^2}{\alpha}$$

$$(\Delta t)_{\text{limit}} = \frac{\Delta x^2 K}{2(h\Delta x + K)\alpha}$$

Data used in simulation :- $N = 5$ (no. of nodes)

$$\Delta x = \frac{L}{N-1} = \frac{0.2}{4} = 0.05 \text{ m}$$

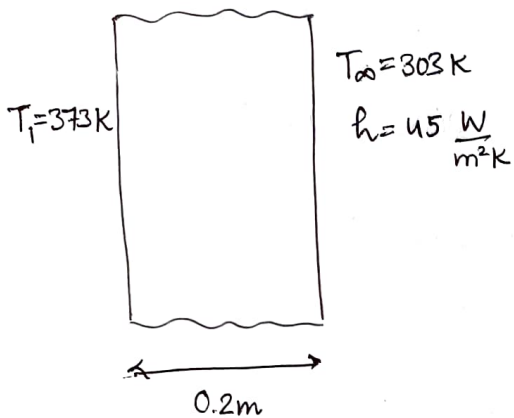
$$(\Delta t)_{\text{limit}} = \frac{(0.05)^2 \times 18}{2(45 \times 0.05 + 18) \times (2.28 \times 10^{-4})} = 4.87329 \text{ seconds.}$$

∴ timestep should be less $(\Delta t)_{\text{limit}}$ to avoid oscillation of final solution.

$$(\Delta t) = 0.01 \times (\Delta t)_{\text{limit}} = 0.0487329 \text{ seconds taken as time step.}$$

(B) $K = K_0(1 + \beta T) = 14(1 + 9.21 \times 10^{-4} T) \left(\frac{W}{mK} \right)$ and $q''' = 30000 \frac{W}{m^3}$

Rest of the problem remains the same.



(i) Analytical Solution:-

→ Governing Differential Equation:-

$$\frac{d}{dx} \left(K \frac{dT}{dx} \right) + q''' = 0$$

$$K \frac{dT}{dx} = -q'''x + C_1$$

$$(1 + \beta T) \frac{dT}{dx} = \frac{-q'''}{K_0} x + \frac{C_1}{K_0}$$

$\underbrace{\frac{C_1}{K_0}}_{C_2}$

$$\int (1 + \beta T) dT = \int \left(\frac{-q'''}{K_0} x + C_2 \right) dx$$

$$T + \frac{\beta}{2} T^2 = -\frac{q'''x^2}{2K_0} + C_2x + C_3 \rightarrow (1)$$

By Boundary (i), at $x=0$, $T_1 = 373K$, $\therefore C_3 = 437.0689$

By Boundary condition (ii), at $x=0.2m$,

$$-K \left(\frac{dT}{dx} \right)_{x=0.2} = h(T_{0.2} - T_\infty)$$

$$-K_0(1 + \beta T) \frac{dT}{dx} \Big|_{x=0.2} = -K_0 \left(\frac{-q'''x}{K_0} + C_2 \right) \Big|_{x=0.2}$$

$$\therefore -K_0 \left(-\frac{q'''x}{K_0} + C_2 \right) = h(T_L - T_\infty)$$

$$C_2 = \frac{q'''L - h(T_L - T_\infty)}{K_0} = \frac{30000 \times 0.2 - 45(T_L - 303)}{14}$$

$$C_2 = 1402.5 - 3.214286 T_L \rightarrow (2)$$

Using (2) in (1) we get

$$T_L + \frac{0.000921}{2} T_L^2 = \frac{-30000 \times (0.2)^2}{2 \times 14} + (1402.5 - 3.214286 T_L) \times 0.2 + 437.0689$$

$$T_L + \frac{0.000921}{2} T_L^2 = -42.85714 + 280.5 + 437.0689 - 0.64286 T_L$$

$$T_L = \frac{-3567.557 \pm \sqrt{(-3567.557)^2 + 4(1965172.117)}}{2}$$

$$T_L = 371.92 \text{ K} \rightarrow (3)$$

∴ Using (3) in (2), we get $C_2 = 207.0427$

$$\therefore 0.0004605 T^2 + T = \frac{-30000 x^2}{2 \times 14} + \frac{207.427}{1} x + \frac{437.0689}{1}$$

$$T^2 + 2171.5527 T = -2326663.56 x^2 + 450438.65 x + 949118.1325$$

$$T = \frac{-2171.5527 \pm \sqrt{(-2171.5527)^2 + 4(-2326663.56 x^2 + 450438.65 x + 949118.13)}}{2}$$

→ (4)

(4) is our analytical temperature distribution.

→ Heat flux ($\dot{q}''$) :- $\dot{q}'' = -k \frac{dT}{dx} \Big|_x = -k_0(1+\beta T) \frac{dT}{dx}$

$$\frac{dT}{dx} \Rightarrow \frac{dT}{dx} + \frac{\beta}{2} \times 2T \times \frac{dT}{dx} = \frac{-\dot{q}'''}{k_0}$$

$$\frac{dT}{dx} = \frac{-\dot{q}''' x + k_0 C_2}{k_0(1+\beta T)}$$

$$\therefore \dot{q}''' = \frac{-k_0(1+\beta T)(-\dot{q}''' x + k_0 C_2)}{k_0(1+\beta T)} = +30000 x - 2898.598$$

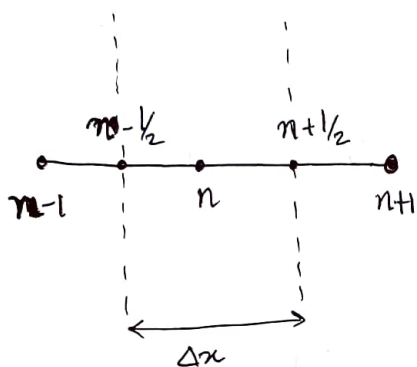
→ (5)

(5) $\dot{q}'' = 30000x - 2898.598$ (Heat flux at any point in the wall)

(i) Numerical Method :- GDE :- $\frac{d}{dx}(k \frac{dT}{dx}) + \dot{q}''' = 0$

(a) for first node 1 :- $T_1 = 373 \text{ K}$ (Temperature Boundary condition) at $x=0$

(b) for interior nodes 2 to $N-1$ (where $N = \text{no. of nodes}$) :-



$$\frac{\left(K(T) \frac{dT}{dx} \right)_{n+1/2} - \left(K(T) \frac{dT}{dx} \right)_{n-1/2}}{\Delta x} + \dot{q}''' = 0$$

$$\frac{K(T_{n+1/2}) \left(\frac{T_{n+1} - T_n}{\Delta x} \right) - K(T_{n-1/2}) \left(\frac{T_n - T_{n-1}}{\Delta x} \right)}{\Delta x} + \dot{q}''' = 0$$

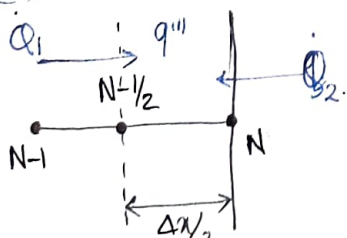
$$K_0 \left(1 + \beta \left(\frac{T_{n+1} + T_n}{2} \right) \right) \left(\frac{T_{n+1} - T_n}{\Delta x^2} \right) - K_0 \left(1 + \beta \left(\frac{T_{n-1} + T_n}{2} \right) \right) \left(\frac{T_n - T_{n-1}}{\Delta x^2} \right) + \dot{q}''' = 0$$

$$\left(T_{n+1} - T_n + \frac{\beta}{2} (T_{n+1}^2 - T_n^2) \right) - \left(T_n - T_{n-1} + \frac{\beta}{2} (T_n^2 - T_{n-1}^2) \right) + \frac{\dot{q}''' \times \Delta x^2}{K_0} = 0$$

$$(T_{n+1} + T_{n-1} - 2T_n) + \frac{\beta}{2} (T_{n+1}^2 + T_{n-1}^2 - 2T_n^2) + \frac{\dot{q}''' \Delta x^2}{K_0} = 0$$

$$T_n = \frac{T_{n+1} + T_{n-1} + \frac{\beta}{2} (T_{n+1}^2 + T_{n-1}^2 - 2T_n^2) + \frac{\dot{q}''' \Delta x^2}{K_0}}{2}$$

(c) For last node ($n=N$)



$$\dot{Q}_1 + \dot{Q}_2 + \dot{q}''' = 0$$

$$-K(T) \times A \times \left. \frac{dT}{dx} \right|_{N-1/2} + hA(T_\infty - T_N) + \dot{q}''' A \frac{\Delta x}{2} = 0$$

$$-k_0 \left(1 + \frac{\beta}{2} (T_{N-1/2} + T_N)\right) \left(\frac{T_{N-1} - T_N}{\Delta x}\right) + \frac{q''' \Delta x}{2} + h(T_\infty - T_N) = 0$$

$$\frac{k_0}{\Delta x} \left(T_{N-1} - T_N + \frac{\beta}{2} (T_{N-1}^2 - T_N^2)\right) + \frac{q''' \Delta x}{2} + h(T_\infty - T_N) = 0$$

$$T_N = \left(\frac{1}{1 + \frac{h \Delta x}{k_0}}\right) \left[T_{N-1} + \frac{\beta}{2} (T_{N-1}^2 - T_N^2) + \frac{q''' \Delta x^2}{2 k_0} + \frac{h \Delta x}{k_0} T_\infty \right]$$

→ Heat flux ($\dot{q}''$) :- (a) for inner nodes $n = 2, 3, 4, \dots, N-1$

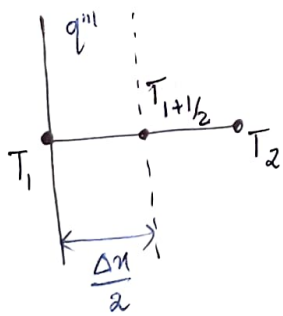
$$-k(T) \left. \frac{dT}{dx} \right|_{n} + q''' \Delta x = \dot{q}''$$

$$-k_0 \left(1 + \beta T_m\right) \left[\frac{T_{n+1/2} - T_{n-1/2}}{\Delta x} \right] + q''' \Delta x = \dot{q}''$$

$$-k_0 \left(1 + \beta T_n\right) \left[\frac{(T_{n+1} + T_n) - (T_n + T_{n-1}))}{2 \Delta x} \right] + q''' \Delta x = \dot{q}''$$

$$-k_0 \left(1 + \beta T_m\right) \left[\frac{T_{n+1} - T_{n-1}}{2 \Delta x} \right] + q''' \Delta x = \dot{q}''$$

(b) for first node ($n=1$) :-



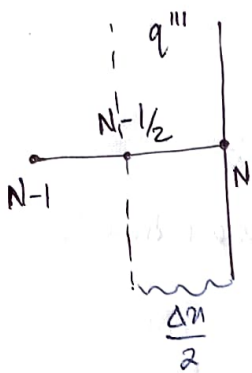
$$\dot{q}'' = -k(T) \left. \frac{dT}{dx} \right|_{n=1} + q''' \frac{\Delta x}{2}$$

$$= -k(T_1) \left[\frac{T_{3/2} - T_1}{\frac{\Delta x}{2}} \right] + \frac{q''' \Delta x}{2}$$

$$= -k(T_1) \left[\frac{\frac{T_2 + T_1}{2} - T_1}{\frac{\Delta x}{2}} \right] + \frac{q''' \Delta x}{2}$$

$$\dot{q}'' = -k_0 \left(1 + \beta T_1\right) \left(\frac{T_2 - T_1}{\Delta x}\right) + q''' \frac{\Delta x}{2}$$

(c) for last node $n=N$,



$$\dot{q}'' = -K(T) \frac{dT}{dx} \Big|_{n=N} + q''' \frac{\Delta x}{2}$$

$$\dot{q}'' = -K_o(1+\beta T_N) \left[\frac{T_N - T_{N-1/2}}{\frac{\Delta x}{2}} \right] + q''' \frac{\Delta x}{2}$$

$$= -K_o(1+\beta T_N) \left[\frac{T_N - \left(\frac{T_N + T_{N-1}}{2} \right)}{\frac{\Delta x}{2}} \right] + q''' \frac{\Delta x}{2}$$

$$\boxed{\dot{q}'' = -K_o(1+\beta T_N) \left[\frac{T_N - T_{N-1}}{\Delta x} \right] + q''' \frac{\Delta x}{2}}$$